



ERTS Lab
Department of Computer Science and Engineering
Indian Institute of Technology Bombay, Powai
Mumbai - 400 076.



Certificate of Participation

This is to certify that **Vineet Sharma**, a student of **SRM Institute of Science and Technology, Tamil Nadu** has participated in the **e-Yantra Robotics Competition (eYRC 2025-26)**.

He/She is a member of the team having the following team members,

1. Nilesh Arumalla
2. Vineet Sharma
3. Ashwin S
4. Mohamed Sheik Abdullah

This team has participated in **MazeSolver Bot** theme.

Prof. Shivaram Kalyanakrishnan

Principal Investigator, e-Yantra
Associate Professor
Department of Computer Science and Engineering
Indian Institute of Technology Bombay



EYRC2025-
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e-Yantra is a project sponsored by MHRD, Government of India, under the National Mission on Education through ICT (NMEICT).

[P.T.O]

eYantra

Engineering a better tomorrow

Theme: *MazeSolver Bot*

Theme Description:

The objective of this project is to design and implement an FPGA-based autonomous MazeSolver Bot (MB) capable of navigating a dynamic warehouse-like maze that simulates a post-harvesting agricultural storage environment. The maze walls resemble stacks of produce sacks and cartons, with layouts that can change to mimic real-world warehouse conditions. Participants are required to build the complete robot from scratch using an FPGA as the core processing unit, designing the system in Verilog HDL. A custom architecture/algorithm must be implemented on the FPGA to handle sensor data acquisition, motor control, decision-making, and wireless communication in parallel. As the MB navigates the arena, it must solve the maze in real time while adapting to blocked or reconfigured paths. Additionally, the robot measures key microclimate parameters such as temperature, relative humidity, and soil moisture at designated zones within the warehouse.

The theme focuses on creating an intelligent FPGA-powered robot that can navigate and monitor agricultural storage environments efficiently.

Theme Learning: *FPGA, C, Verilog, CPU Architecture, Build-a-Bot, Parallel Processing, Serial Communication*

Certificate Level: The team has been awarded **Level 3** certificate in the e-Yantra Robotics Competition (eYRC).

Level 3 indicates: *The team showed a fair understanding of the problem statement by implementing a partial solution of the problem statement.*

Certificate Level Matrix

Certificate Level	Certificate Description
Level 1	Certificate of Merit
Level 2	Certificate of Completion
Level 3	Certificate of Participation
Level 4	Letter of Participation

* Level 1 > Level 2 > Level 3 > Level 4 (Level 1 is highest achievement)